

5(a)	<i>closed</i> blood flows through, (blood) vessels; A three of heart, arteries, veins, capillaries <i>double</i> blood flows through the heart twice in one complete circulation (of the body) / AW ; A ref. to pulmonary and systemic circuits / to lungs and rest of body	2
5(b)(i)	pulmonary vein ; semi-lunar / AW, valve ; A pulmonary valve R aortic valve right, atrium / auricle ;	3

Question	Answer	Marks
5(b)(ii)	<i>any three from:</i> 1 left ventricle / chamber Y, pumps blood into, systemic circulation / described or right ventricle / chamber X pumps blood into, pulmonary circulation / described or distance travelled by blood in systemic circulation is greater than distance travelled by blood in pulmonary circulation / AW; 2 to overcome great(er) resistance to flow in systemic circulation ; ora 3 high (blood) pressure is required for blood to travel around the systemic circulation ; 4 high pressure requires more muscular force ; ora 5 pulmonary capillaries, rupture easily / damaged by high pressure ; 1 more cardiac muscle	3
5(c)	<i>any four from:</i> 1 impulse / wave of excitation / AW, passes from SAN to atria (muscles) ; R nervous impulse / signal <i>once only</i> 2 atria both contract, together / at the same time ; A atrial systole <i>if not contradicted by one contracting before the other</i> 3 atria contract before ventricles ; 4 fibrous / non-conducting, tissue prevents impulse travelling to ventricles ; 5 impulse delayed at AVN ; 6 AVN passes impulse to, bundle of His / Purkyne fibres ; 7 Purkyne fibres conduct impulses to <u>muscle</u> in wall of ventricles ; 8 ventricles contract together (if mp2 not awarded) ; A ventricular systole if not contradicted as for atria 9 ventricles contract from the bottom upwards ;	4